	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{K}_{0^+}^{\#1}\dagger$	$\frac{1}{2}(-3\,\Upsilon_1\,k^2-3\,\overset{(2)}{\kappa}_3)$	0
$\mathcal{K}_{0^-}^{\#1}\dagger^{\alpha\beta\chi}$	0	0

	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{1^-}^{\#1}{}_{\alpha}$	$\mathcal{K}_{1^-}^{\#2}{}_{\alpha}$
$\mathcal{K}_{1^+}^{\#1}\dagger^{\alpha\beta}$	$-k^2\,\overset{(4)}{\kappa}_3$	$-\frac{k^2\,\overset{(4)}{\kappa}_3}{\sqrt{2}}$	0	0
$\mathcal{K}_{1^+}^{\#2}\dagger^{\alpha\beta}$	$-\frac{k^2\,\overset{(4)}{\kappa}_3}{\sqrt{2}}$	$-\frac{k^2\,\overset{(4)}{\kappa}_3}{2}$	0	0
$\mathcal{K}_{1^-}^{\#1}\dagger^{\alpha}$	0	0	$-k^2\,\overset{(4)}{\kappa}_2-\overset{(2)}{\kappa}_3$	$\frac{k^2\,\overset{(4)}{\kappa}_2+\overset{(2)}{\kappa}_3}{\sqrt{2}}$
$\mathcal{K}_{1^-}^{\#2}\dagger^{\alpha}$	0	0	$\frac{k^2\,\overset{(4)}{\kappa}_2+\overset{(2)}{\kappa}_3}{\sqrt{2}}$	$\frac{1}{2}(-k^2\,\overset{(4)}{\kappa}_2-\overset{(2)}{\kappa}_3)$

	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{K}_{2^+}^{\#1}\dagger^{\alpha\beta}$	0	0
$\mathcal{K}_{2^-}^{\#1}\dagger^{\alpha\beta\chi}$	0	0

	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{J}_{0^+}^{\#1}\dagger$	$\frac{1}{\text{Det}(0^+)}$	0
$\mathcal{J}_{0^-}^{\#1}\dagger^{\alpha\beta\chi}$	0	0

	$\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{J}_{1^-}^{\#1}{}_{\alpha}$	$\mathcal{J}_{1^-}^{\#2}{}_{\alpha}$
$\mathcal{J}_{1^+}^{\#1}\dagger^{\alpha\beta}$	$-\frac{4}{9\,k^2\,\overset{(4)}{\kappa}_3}$	$-\frac{2\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_3}$	0	0
$\mathcal{J}_{1^+}^{\#2}\dagger^{\alpha\beta}$	$-\frac{2\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_3}$	$-\frac{2}{9\,k^2\,\overset{(4)}{\kappa}_3}$	0	0
$\mathcal{J}_{1^-}^{\#1}\dagger^{\alpha}$	0	0	$\frac{2}{3\,\text{Det}(1^-)}$	$-\frac{\sqrt{2}}{3\,\text{Det}(1^-)}$
$\mathcal{J}_{1^-}^{\#2}\dagger^{\alpha}$	0	0	$-\frac{\sqrt{2}}{3\,\text{Det}(1^-)}$	$\frac{1}{3\,\text{Det}(1^-)}$

	$\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{2^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{J}_{2^+}^{\#1}\dagger^{\alpha\beta}$	0	0
$\mathcal{J}_{2^-}^{\#1}\dagger^{\alpha\beta\chi}$	0	0

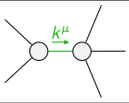
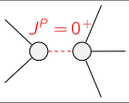
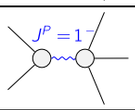
Abbreviations used in matrices

$$\Upsilon_1 == \overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2 \;\&\&\; \text{Det}(0^+) == -\frac{3}{2}(k^2(\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2)+\overset{(2)}{\kappa}_3) \;\&\&\; \text{Det}(1^-) == -\frac{3}{2}(k^2\,\overset{(4)}{\kappa}_2+\overset{(2)}{\kappa}_3)$$

Lagrangian

$$\overset{(2)}{\kappa}_3\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}^{\chi}{}_{\beta\chi}+\overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+\overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+$$
$$\overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta}-2\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta}-\overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta}-\frac{1}{2}\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{0^-}^{\#1}{}^{\alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta}+\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi}+\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta}+\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta}+\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta}+\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta}+\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta}+\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi}+\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta}+\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi}+\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}_{1^-}^{\#1}{}^{\alpha}+2\,\mathcal{J}_{1^-}^{\#2}{}^{\alpha} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi}+\partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}_{1^+}^{\#1}{}^{\alpha\beta} == \mathcal{J}_{1^+}^{\#2}{}^{\alpha\beta}$	3	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi}+2\,\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$
$\mathcal{J}_{2^-}^{\#1}{}^{\alpha\beta\chi} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta}+$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} ==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta}+$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
$\mathcal{J}_{2^+}^{\#1}{}^{\alpha\beta} == 0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta}+3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta}+3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	17	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{1}{\overset{(4)}{\kappa}_3}$
	1	$-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2}$	$-\frac{2}{3(\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2)}$
	3	$-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_2}$	$\frac{2}{3\,\overset{(4)}{\kappa}_2}$
Resolved unitarity condition(s):	(Demonstrably impossible)		